

ACRBP388

## L-Methionine

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>L-蛋氨酸</b><br><b>L-Methionine</b>                                                                                                                                                                                                                                |
| <b>Cat No. :</b>                                      | <b>BP388-100</b>                                                                                                                                                                                                                                                   |
| <b>Synonyms</b>                                       | L-2-Amino-4-(methylthio)butyric acid; Met                                                                                                                                                                                                                          |
| <b>CAS No</b>                                         | 63-68-3                                                                                                                                                                                                                                                            |
| <b>Molecular Formula</b>                              | C5 H11 N O2 S                                                                                                                                                                                                                                                      |
| <b>Supplier</b>                                       | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                              |                            |                       |
|----------------------------------------------------------------------------------------------|----------------------------|-----------------------|
| <b>Physical State</b><br>Solid                                                               | <b>Appearance</b><br>White | <b>Odor</b><br>Slight |
| <b>Emergency Overview</b><br>Air sensitive. May form combustible dust concentrations in air. |                            |                       |

#### Classification of the substance or mixture

Based on available data, the classification criteria are not met

#### Label Elements

None required

#### **Physical and Chemical Hazards**

None identified. May form combustible dust concentrations in air.

#### **Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### **Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely

be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

May form explosible dust-air mixture if dispersed. This product does not contain any known or suspected endocrine disruptors.

### SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component  | CAS No  | Weight % |
|------------|---------|----------|
| Methionine | 63-68-3 | 98       |

### SECTION 4. FIRST AID MEASURES

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.

#### Ingestion

Do NOT induce vomiting. Get medical attention.

#### Most important symptoms and effects

No information available.

#### Self-Protection of the First Aider

No special precautions required.

#### Notes to Physician

Treat symptomatically.

### SECTION 5. FIRE-FIGHTING MEASURES

#### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

#### Specific Hazards Arising from the Chemical

Dust can form an explosive mixture with air. Fine dust dispersed in air may ignite.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

#### Personal Precautions

Ensure adequate ventilation.

#### Environmental Precautions

See Section 12 for additional Ecological Information.

## L-Methionine

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin and eyes. Do not breathe dust. Minimize dust generation and accumulation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

None under normal use conditions.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene       |                                   |                 |             |                       |
| Natural rubber |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                   |                                          |
|------------------------------------------------|-----------------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                             |                                          |
| <b>Physical State</b>                          | Solid                             |                                          |
| <b>Odor</b>                                    | Slight                            |                                          |
| <b>Odor Threshold</b>                          | No data available                 |                                          |
| <b>pH</b>                                      | 5.5-6.5                           |                                          |
| <b>Melting Point/Range</b>                     | 276 - 279 °C / 528.8 - 534.2 °F   |                                          |
| <b>Softening Point</b>                         | No data available                 |                                          |
| <b>Boiling Point/Range</b>                     | No information available          |                                          |
| <b>Flash Point</b>                             | No information available          | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                    | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available          |                                          |
| <b>Explosion Limits</b>                        | No data available                 |                                          |
| <b>Vapor Pressure</b>                          | No data available                 |                                          |
| <b>Vapor Density</b>                           | Not applicable                    | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                 |                                          |
| <b>Bulk Density</b>                            | No data available                 |                                          |
| <b>Water Solubility</b>                        | SOLUBLE IN WATER: 53.7 G/L (20°C) |                                          |
| <b>Solubility in other solvents</b>            | No information available          |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                   |                                          |
| <b>Component</b>                               | <b>log Pow</b>                    |                                          |
| Methionine                                     | -1.87                             |                                          |
| <b>Autoignition Temperature</b>                | No data available                 |                                          |
| <b>Decomposition Temperature</b>               | No data available                 |                                          |
| <b>Viscosity</b>                               | Not applicable                    | Solid                                    |
| <b>Explosive Properties</b>                    | No information available          |                                          |
| <b>Oxidizing Properties</b>                    | No information available          |                                          |
| <b>Molecular Formula</b>                       | C5 H11 N O2 S                     |                                          |
| <b>Molecular Weight</b>                        | 149.21                            |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                |
|---------------------------------|------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions. Air sensitive. |
| <b>Hazardous Reactions</b>      | No information available.                      |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.       |
| <b>Conditions to Avoid</b>      | Incompatible products.                         |
| <b>Materials to avoid</b>       | Strong oxidizing agents.                       |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

**Product Information**

## SAFETY DATA SHEET

L-Methionine

## (a) acute toxicity;

| Component  | LD50 Oral              | LD50 Dermal | LC50 Inhalation |
|------------|------------------------|-------------|-----------------|
| Methionine | LD50 = 36 g/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

## (d) respiratory or skin sensitization;

Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects .

## Persistence and Degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.

## Bioaccumulative Potential

Bioaccumulation is unlikely

| Component  | log Pow | Bioconcentration factor (BCF) |
|------------|---------|-------------------------------|
| Methionine | -1.87   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## SAFETY DATA SHEET

## L-Methionine

**Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component  | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Methionine | -                                                   | -                                       | X    | X     | 200-562-9 | X    | X   | X     | X    | X    | X    | KE-01485 |

**National Regulations****SECTION 16. OTHER INFORMATION****Creation Date**

12-Sep-2014

**Revision Date**

20-Apr-2024

**Revision Summary**

Not applicable.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

## L-Methionine

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**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**